ECDLP transfer methodology: known-case and net-cost validation

aburan28/crypto research record

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Outcome

The fail-closed transfer methodology and native validator classify six known public cases consistently with their frozen source evidence. All six assessments pass the version-three machine contract, and ten semantic controls pass, including rejection of unnamed solvers, stage-only promotions, unpriced work, inconsistent phase totals, unsupported measurements, and incorrect net ratios. Two fixtures support a valid subgroup transfer, two remain unknown at the transfer gate, and two are refuted. None supports a modeled or measured computational advantage.

The extension now defines what "a weaker destination" means operationally. A candidate must name the destination solver, show that it reaches the transported order-`r` component, and compare three arms in one unit: the strongest direct source solver, a generic solver on the destination image, and the specialized destination pipeline with all nine transport/solve/recovery phases charged. A destination-stage gain is recorded but cannot promote the net-transfer verdict.

The replay also found and repaired a pre-existing build defect in `tools/ecdlp-surfaces`: the checked-in binary module was not exported from the crate root, so the package's own validation script could not compile the n=83 checker. After adding the missing module export, the existing package reported 22 tests passed, zero failed, Clippy clean, and byte-for-byte replay of its committed examples.

Requested scope and evidence

Requirement	Status	Evidence
Put the methodology in `aburan28/crypto`	Verified complete	`docs/transfer-assessment/METHODOLOGY.md`
Make it machine-checkable	Verified complete	`tools/transfer-assessment`, v3 schema, template, ten semantic controls
Run it on known transfers	Verified complete for six bounded public cases	`assessments/` and `validation-summary.json`
Include covers, surfaces, higher genus, and lattices	Verified complete as diagnostic coverage	quartic image, product surfaces, genus-3 cover, torsion-free lattice control
Define when to use the repository skills	Verified complete	`docs/transfer-assessment/MEASUREMENT.md` and the `ecdlp-surfaces` skill handoff
Measure whether a destination solver is actually weaker	Protocol and machine gates complete; no new solver run performed	named-solver record, three arms, nine phases, net ratio, replay/interval/scaling gates
Establish a new faster ECDLP reduction	Not established and not implied	all advantage verdicts are `unknown` or `not_applicable`
Validate arbitrary covers/lattices/surfaces	Open beyond the six cases	each new candidate still needs its own identities, map, subgroup, solver, and cost evidence

The validation profile is an assessment consistency check. It does not make an official state transition, independently prove every cited artifact, or turn template completion into evidence.

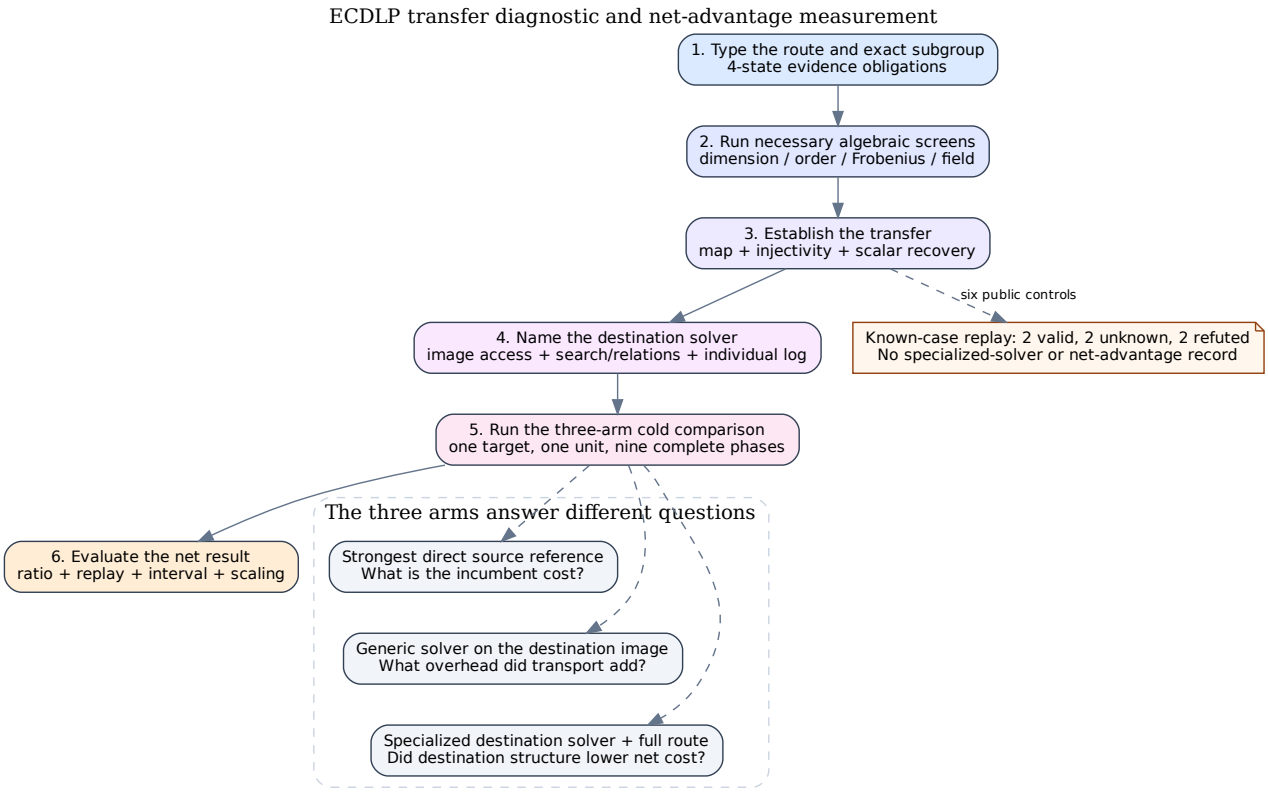


Figure 1: The diagnostic first establishes the transfer and named solver, then admits only a three-arm, cold end-to-end net comparison.

Pinned basis

Repository revision: `3d4a891bd07000cf63935f5c876a22ff5e193315`.

The cases cite these exact paths at that revision:

- tools/ecdlp-surfaces/examples/transfer_certificate.json;
- tools/ecdlp-surfaces/examples/frobenius_result.json;
- tools/ecdlp-surfaces/examples/binary_n83_a0.json;
- tools/ecdlp-surfaces/README.md; and
- research/hyperelliptic_cover_infrastructure_20261006/README.md.

The lattice case is a known-false type control from the methodology: an ordinary additive lattice `Z^m` is torsion-free, so a homomorphism from the finite cyclic group `Z/rZ` has zero image. It does not reject finite lattice quotients, tori, endomorphism-kernel lattices, or relation lattices; those must be typed and audited separately.

Exact classifications

Case	Strongest established stage	Valid transfer	Algorithmically relevant	Modeled / measured advantage
Small-prime quartic image	explicit coordinate isomorphism, subgroup and recovery checks	supported	supported for generic group algorithms	unknown / unknown
n=83 inclusion `E_0 -> E_0 x E_1`	explicit factor inclusion and projection recovery	supported	supported for generic group algorithms	unknown / unknown
Compatible degree-four Frobenius polynomial	necessary isogeny-class filter only	unknown	unknown	unknown / unknown
n=83 genus-3 cover	curve construction and `C -> E` map; exact n=83 Jacobian pullback unexecuted	unknown	unknown	unknown / unknown
n=83 proposed `E_0 -> E_1 x E_1`	order and Frobenius obstruction	refuted	refuted	not applicable
proposed `Z/rZ -> Z^m`	torsion obstruction	refuted	refuted	not applicable

This is the intended discrimination:

1. A faithful representation may be a valid transfer without an advantage.
2. Frobenius compatibility is necessary evidence, not a homomorphism.
3. A curve map `C -> E` is not the required executable subgroup map `E -> J(C)`; the n=83 pullback remains unknown until replayed.
4. Group-order, Frobenius, and torsion contradictions reject impossible paths before solver work.
5. Generic rho is applicable to both valid image subgroups, but that says only that the DLP survives transport; it does not make either destination weaker.

Net-advantage decision rule

The measurement record contains three complete arms: `source_reference`, `destination_generic_control`, and `destination_candidate`. Its primary scope is one cold target. Candidate cost is the sum of route search, construction, forward map, solver setup, relation/walk, linear algebra, individual-log/descent, source recovery, and verification. The only net ratio eligible for promotion is `source_reference.total / destination_candidate.total`.

The destination generic control answers whether any observed change is merely representation overhead. The specialized arm answers whether index calculus, Weil descent, or another named method exploits the destination. Both remain secondary to the net ratio. Amortization, online-only intervals, and individual phase ratios are recorded separately and cannot promote the cold verdict.

The skill handoff is deliberate: `transfer` and `ecdlp-surfaces` find and certify routes; `curve-algebra` establishes relevant endomorphism/Frobenius actions; `ecbench-measure` runs the matched cross-method comparison; and `ecbench-bounds` tests an admitted measurement against the established frontier. `report-evidence` and `research-visuals` govern every run and result.

Native validator and controls

`tools/transfer-assessment` parses every assessment, binds each supported or refuted state and every cost arm to a declared source id, and enforces cumulative verdict gates:

- `valid_transfer=supported` requires supported map correctness, subgroup preservation, and scalar recovery;
- `algorithmically_relevant=supported` also requires a named solver with supported applicability, image access, search/relation compatibility, and individual-log/recovery gates;
- modeled advantage additionally requires the cold one-target scope, all three arms, a common unit, nine-phase totals, no unpriced work, a matching ratio greater than one, correctness, resources, and bound evidence;
- measured advantage additionally requires measured arms, five verified pairs, four sizes, a replay receipt, and a 95% interval entirely above one;
- novelty cannot be promoted without the earlier cumulative verdicts; and
- the diagnostic profile refuses an official status change.

Ten Rust tests cover a valid unknown record, a supported transfer, missing transfer gates, unbound evidence, missing solver details, a complete modeled control, stage-only and unpriced rejection, inconsistent phase/ratio arithmetic, a complete measured control, and missing measured-evidence gates. Result: 10 passed, 0 failed.

The six study assessments then produced `transfer-assessment-validation-summary-v1` with `assessments_checked=6`, `all_valid=true`, and no validation errors.

Reproduction

Environment:

```
Linux 6.18.44 x86_64 GNU/Linux
rustc 1.99.0 (b940084d7 2026-09-28)
cargo 1.99.0 (5f94df478 2026-08-27)
Graphviz 2.42.4
Pandoc 3.1.11.1
```

Commands:

```
cargo test --release --locked \
  --manifest-path tools/transfer-assessment/Cargo.toml

cargo clippy --release --locked \
  --manifest-path tools/transfer-assessment/Cargo.toml -- -D warnings

npx --yes ajv-cli@5.0.0 validate --spec=draft2020 \
  -s docs/transfer-assessment/assessment.schema.json \
  -d docs/transfer-assessment/assessment-template.json \
  -d 'research/transfer-methodology-validation-20261013/assessments/*.json'
```

```
cargo run --release --locked \
  --manifest-path tools/transfer-assessment/Cargo.toml -- \
  --output research/transfer-methodology-validation-20261013/validation-summary.json \
  research/transfer-methodology-validation-20261013/assessments/*.json

tools/ecdlp-surfaces/check.sh

cargo test --release --lib --quiet
```

The first surface-package replay at the pinned base failed during compilation:

`error[E0432]: unresolved import ` ecdlp_surfaces::binary `

After `pub mod binary;` was restored in `tools/ecdlp-surfaces/src/lib.rs`, the same command passed 9 binary tests and 13 math tests, Clippy, and committed example regeneration. The repair changes module reachability only; the fixture mathematics and frozen JSON were not edited.

The final repository-wide run on the solver-aware extension, with all compile-time fixtures materialized and the repository's local-socket tests permitted, reported 4,058 passed, zero failed, and 114 ignored in 145.78 seconds. The first sparse/sandboxed attempt from the original methodology run reported 4,043 passed and 15 failed: six failures were missing sparse-checkout fixtures, and nine were local-socket `Operation not permitted` errors. After materializing the exact two remaining fixture directories, the six affected tests passed; all 37 `pollard_collab` tests passed with local sockets enabled; then the clean full run above passed. These were access/setup failures, not mathematical evidence and not changes to the tested implementation.

Graph scope

The boundary ledger, index-calculus scoreboard, progress timeline, curve registry, cover catalog, and existing surface diagrams were checked for applicability. None changes: this PR adds no new curve identity, map edge, admitted performance measurement, solver advantage, or official claim. The study-local diagram records only diagnostic classifications.

Limits and next discriminating check

This round validates methodology behavior on bounded known cases, not general completeness. It contains no performance benchmark, so no `benchmarks/` entry or PNG performance chart is required. No new or substantially revised mathematical theorem is asserted, so there is no new Lean theorem to map.

The next higher-genus workstream has three ordered gates:

1. Execute the native n=83 `OrdinaryBinaryCover` pullbacks for the exact pinned `P` and `Q`; verify image order `r`, degree-three pull/push composition, exceptional fibers, and source scalar recovery.
2. Select and implement a destination solver adapter that can reach that exact Jacobian image. Before n=83 performance work, validate factor-base, relation, linear-algebra, individual-log, norm/projection, and false-relation controls on bounded public examples.
3. Run the three-arm cold comparison over at least four faithful toy sizes, preserve each phase and failure, then apply the relevant n=83 family gate. A stage crossover is not a method crossover; an n=83 result is not an automatic n=131 result.

Completing gate one would support the transfer, not an advantage. Completing gate two would support solver applicability, not net speed. Only gate three can populate a measured advantage record.

Conductor scope coordination could not be performed: `conductor` was absent from this execution environment and the required command exited 127. Work was kept to `docs/transfer-assessment/`, `tools/transfer-assessment/`, this study directory, the `ecdlp-surfaces` skill handoff, and the one-line surface module repair disclosed above.